



Lab: Create an EDA Blueprint

Estimated time: 30 minutes

Learning objectives

After completing this lab, you will be able to:

- Design a scalable, secure, and efficient data architecture blueprint to optimize supply chain operations
- Enable effective decision-making and ensure compliance through tailored data architecture solutions

Prerequisites:

You can complete your tasks using any text editor, such as Word, Google Docs, or Notepad.

- If you install Microsoft Word, you can proceed directly with the instructions. However, if you don't have Microsoft Word, you can sign up for a free version of Microsoft 365 online. Follow [these instructions](#) to create your account.
- You can also use Word Online with a Microsoft 365 subscription or access through a free account. If you're new to Word and would like to learn how to get started, you can find helpful resources here: [Getting Started with Word Online](#).
- For this lab, you can use [Lucidchart](#). You only need the free version of LucidChart. If you choose to purchase a paid plan, you will be responsible for any associated charges. If you've not yet signed up for the free account, you can follow the [instructions](#) to get started. You may alternatively use applications like [Microsoft Visio](#).

Introduction

Reliable and accurate data is the foundation of effective strategy development and decision-making in today's data-driven landscape.

Enterprise Data Architecture (EDA) provides a structured framework for collecting, storing, integrating, and managing data. In a supply chain company, a well-designed EDA ensures:

- Seamless data flow across systems
- Improved data consistency and accessibility
- Maintained data quality

By adopting a structured EDA approach, supply chain companies can enhance operations, make informed decisions, and stay competitive in a rapidly evolving market.

Scenario overview

A supply chain company aims to streamline operations and improve decision-making by developing an Enterprise Data Architecture (EDA) blueprint. The process begins with identifying key data sources, such as inventory systems, transportation logs, supplier databases, and customer orders. The team defines data flow processes, integrating real-time tracking and analytics platforms. Using cloud-based solutions, they establish a centralized data lake for seamless storage and retrieval. The blueprint outlines data governance policies, ensuring security and compliance. Advanced analytics tools are incorporated to forecast demand, optimize routes, and reduce costs. This comprehensive EDA blueprint becomes the foundation for the company's digital transformation journey.

Exercise: Create an Enterprise Data Architecture blueprint for a supply chain company

In this lab, you will design an enterprise data architecture blueprint for a supply chain company that provides raw materials for garment manufacturing. The blueprint will focus on core components: data sources, ingestion, storage, processing, analytics, and governance.

You will be guided step by step to design a framework that ensures efficient data flow, scalability, and compliance. By the end of this lab, you will have a comprehensive and structured data architecture plan tailored to streamline supply chain operations.

Template to use

If you are using Microsoft Word, right-click the link and download the [template](#) document. You can use the template to perform the tasks outlined in the lab.

You can also use the [PDF format](#) for any other word-processing software.

Task 1: Understand business requirements: Identify the specific data requirements for optimizing supply chain operations.

Step 1: Compile a list of all critical data sources

Example

Key data resources
Supplier database

Step 2: Define practical applications for the data

Example

Critical use cases
Inventory forecasting

Step 3: Outline the data compliance standards and regulations that must be followed

Data compliance requirements	Description
GDPR	• Applies if the company handles personal data of EU citizens • Ensure data minimization, lawful processing, and user consent • Provide mechanisms for data access, rectification, and deletion

Task 2: Identify and define the core components of data architecture

Step 1: Data sources: Identify and categorize all internal and external data sources

Example:

Data source type	Examples
Internal	ERP systems, warehouse management systems, production logs
External	Supplier APIs, weather data, market trends

Step 2: Data ingestion: Establish the data ingestion methods.

Example

Ingestion method	Description	Examples	Tools
Batch ingestion	Processes data at scheduled intervals	Daily supplier reports	Apache Kafka, AWS Kinesis, Talend

Step 3: Data storage: Define the data storage types

Example

Storage type	Purpose	Examples
Data Lake	Stores raw, unprocessed data	Hadoop, AWS S3

Step 4: Data processing: Define the data processing methods

Example:

Processing method	Description	Examples
ETL/ELT pipelines	Transform raw data for further analysis	Apache Spark, Talend, AWS Glue, Microsoft Azure Data Factory

Step 5: Analytics: Identify and report business intelligence capabilities

Example:

Analytics type	Description	Examples/Use cases	Tools
Dashboards	Visualize key metrics such as inventory levels, supplier performance, and cost analysis	Inventory levels, supplier performance, cost analysis	Tableau, Power BI

Step 6: Data governance: Define the data governance tasks.

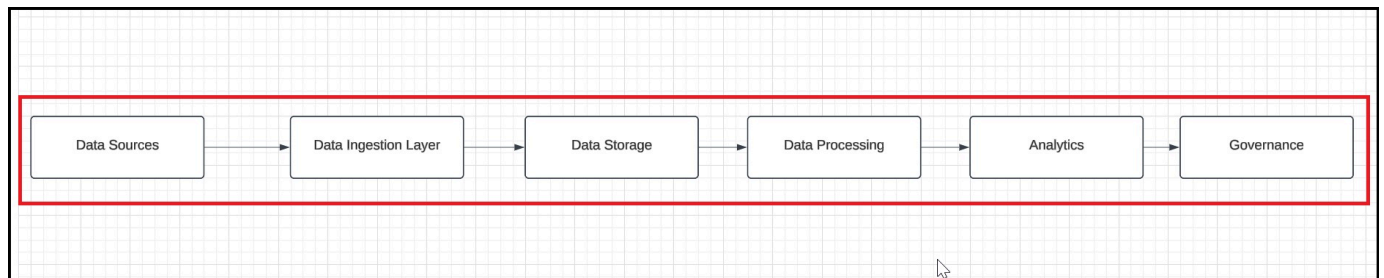
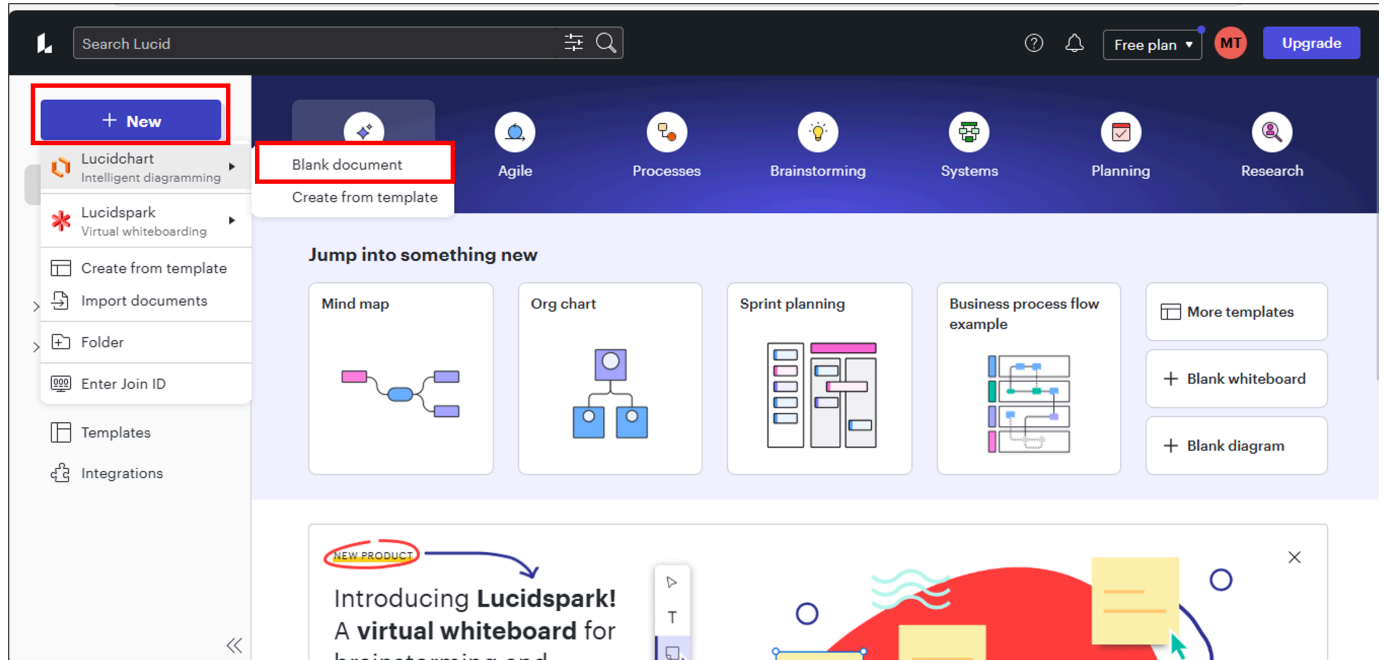
Example:

Governance tasks	Description	Examples/Tools
Data ownership & stewardship	Define roles and responsibilities for data management	Collibra, Alation

Task 3: Create the blueprint using Lucidchart

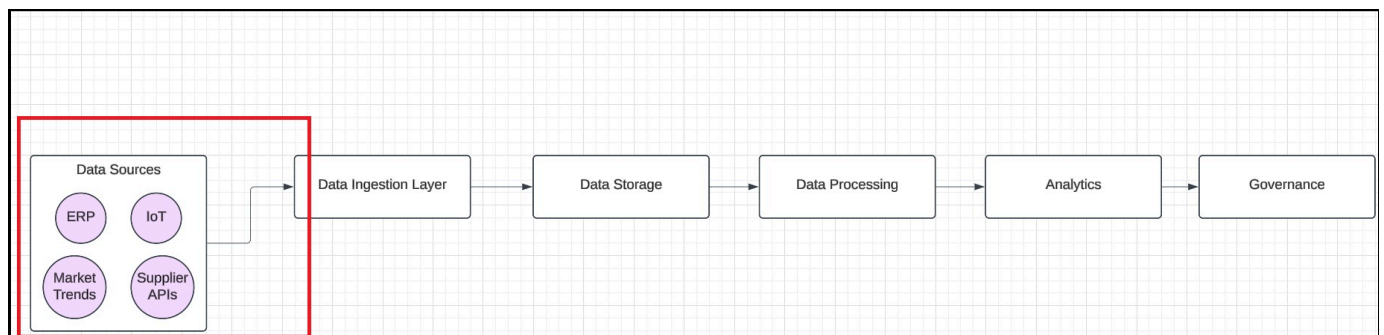
Step 1: Set-up the core layers

Open Lucidchart and click the **New** button, then select **Blank Document** to create a new diagram. Add six rectangular shapes, each representing a core section: **Data Sources**, **Data Ingestion**, **Data Storage**, **Data Processing**, **Analytics**, and **Governance**. Finally, connect these sections with arrows to illustrate the data flow between them.



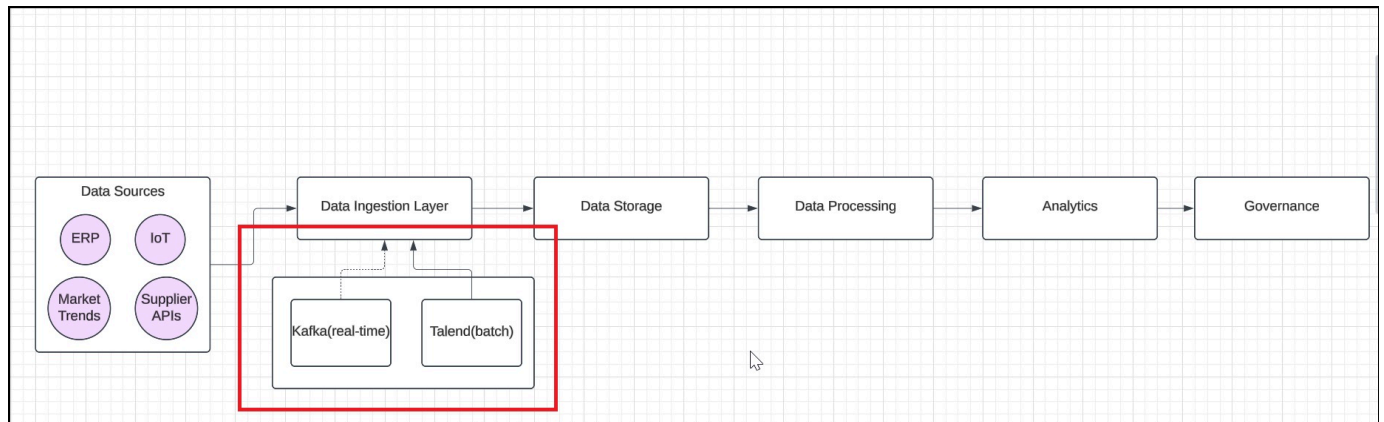
Step 2: Populate the data-sources

Drag and drop circular shapes to represent data sources (e.g., **Supplier APIs**, **ERP system**, **IoT devices**, **Market Trends API**) and connect them to the **Data Ingestion Layer** with arrows.



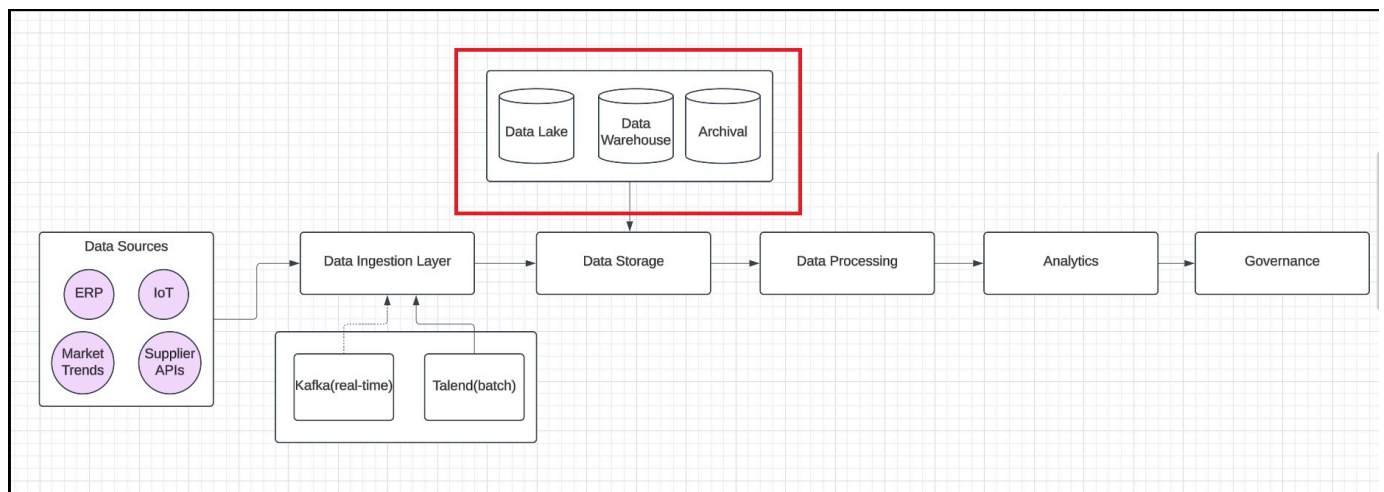
Step 3: Design data ingestion

List the tools and mechanisms (e.g., **Kafka for real-time, Talend for batch**) under the rectangle labeled **Data Ingestion Layer** and annotate the arrows to specify batch or real-time ingestion.



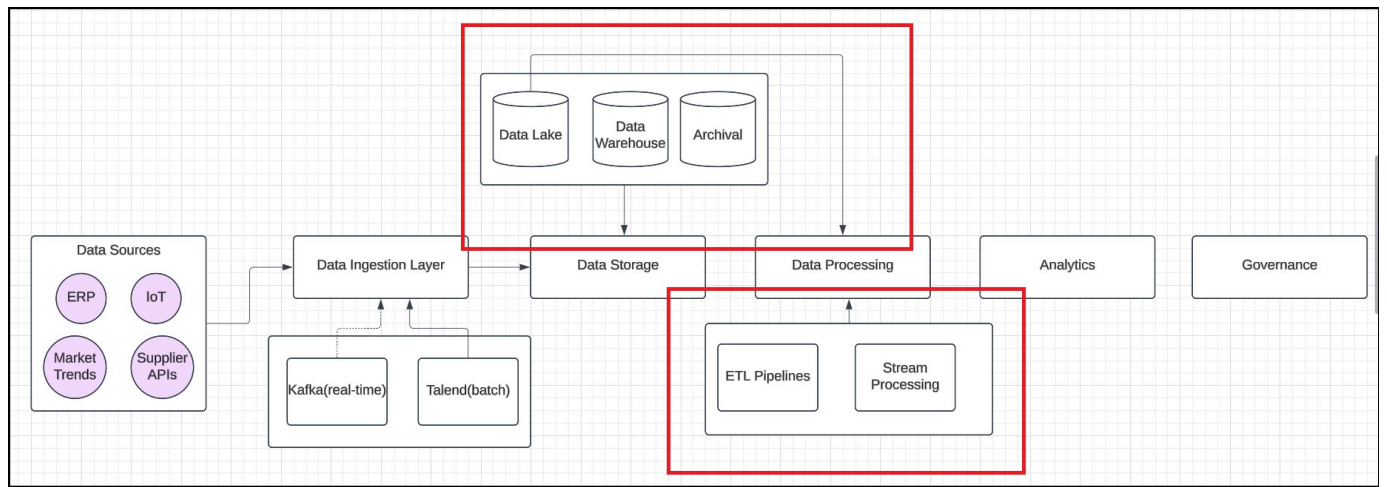
Step 4: Define the storage architecture

Add cylinder shapes to represent each storage tier: Raw Data for the Data Lake, Structured Data for the Data Warehouse, and Archival Storage. Then, establish connections from all storage tiers to the **Data Storage** box.



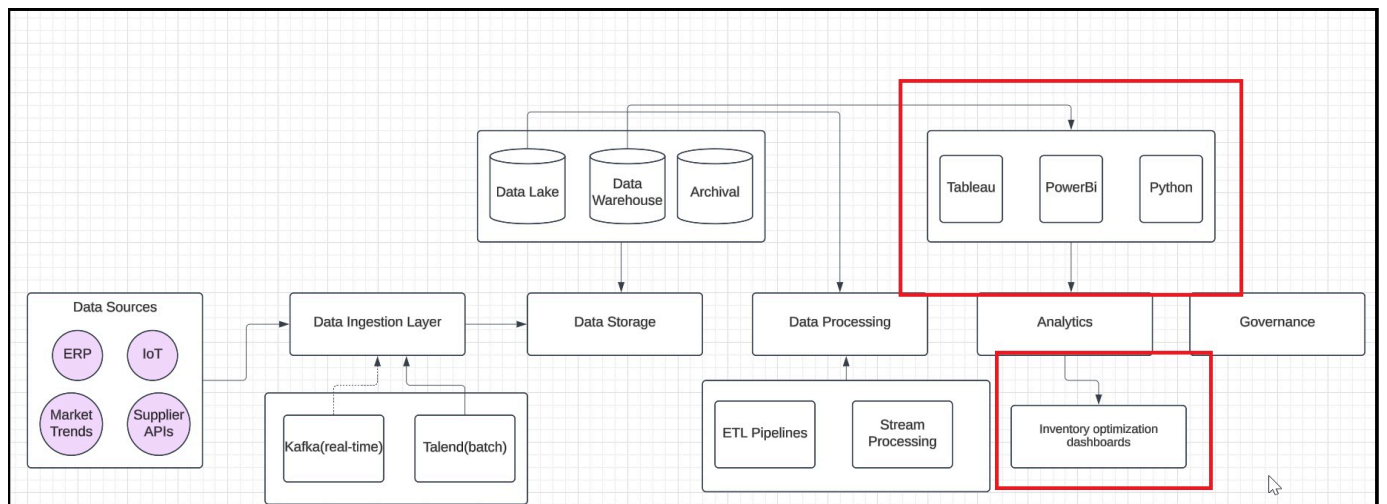
Step 5: Detail data-processing

Under the **Data Processing Layer** rectangle, list the processing methods, such as **ETL pipelines for data cleaning and transformation** and **stream processing for real-time analysis**. Then, connect the **data lake** to the **processing layer** to illustrate the data flow.



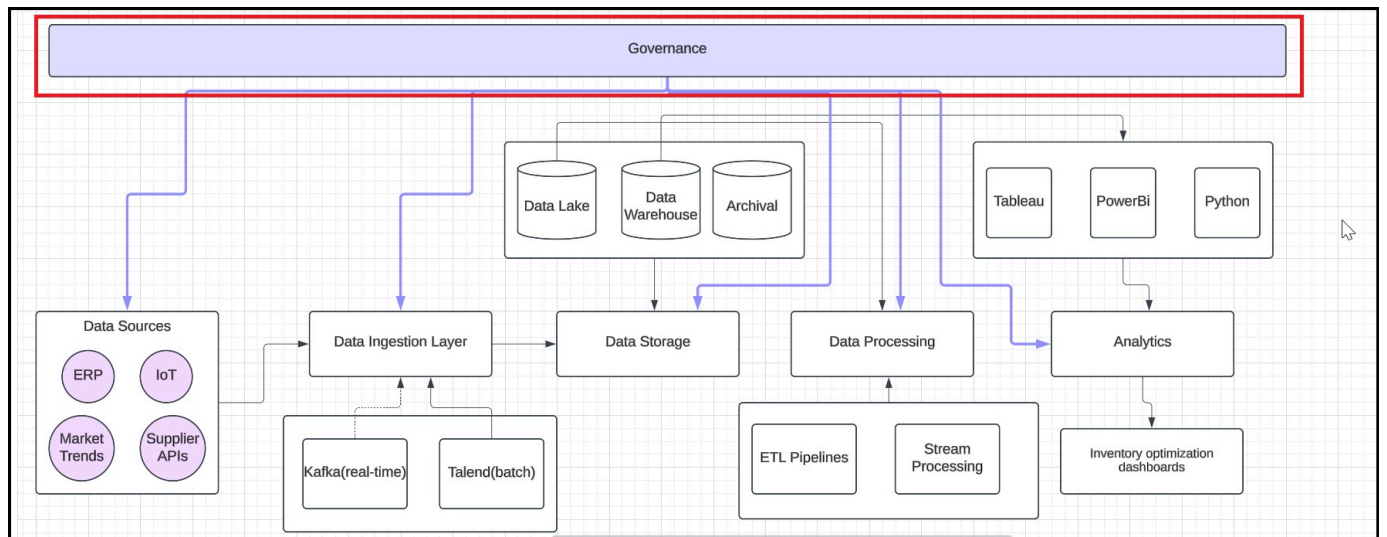
Step 6: Map the analytics

Create rectangular shapes to represent **analytics tools**, such as **Tableau**, **Power BI**, and **Python scripts**. Next, establish connections between the **data warehouse** and these analytics tools to illustrate how data flows for analysis. Finally, include an example output, such as an **inventory optimization dashboard**, to demonstrate the insights generated from the data.



Step 7: Illustrate data governance

Add a rectangle labeled **Data Governance** at the top and connect it to all layers.



Step 8: Finalize and share the blueprint

Export the **blueprint** in **PDF** or **PNG format** to ensure clarity and accessibility. Then, integrate the exported file into the **documentation** to support the implementation process, providing stakeholders with a clear visual reference.

Summary

Congratulations on completing the hands-on lab: Create an EDA Blueprint. In this lab, you explored how to design an enterprise data architecture blueprint for a supply chain company.

Here is the [exemplar](#) document for your reference.

Right-click to open the file. Please remember that the exemplar is just a point of view and not a definitive solution.